

# JOBPILOT AI

## AI POWERED JOB SEARCH PLATFORM

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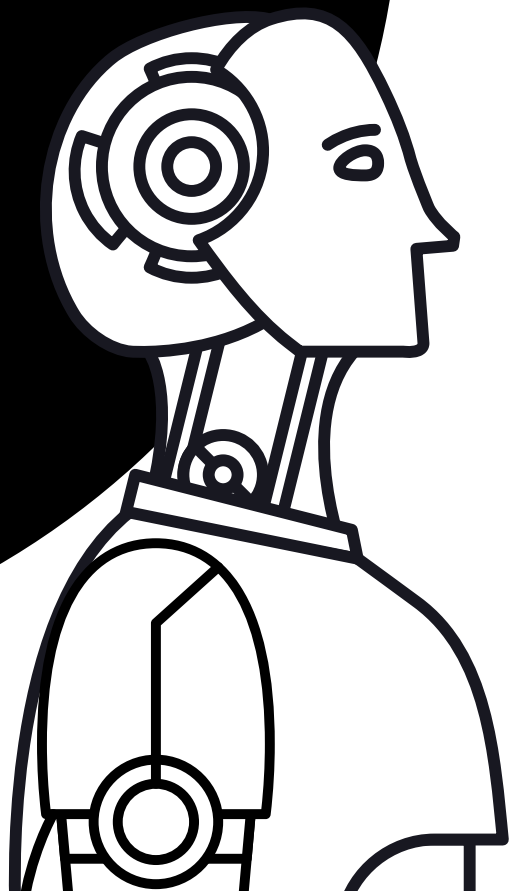
SUPERVISOR - Dr S Poonkuntran

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Team Members :-

|                   |              |
|-------------------|--------------|
| Ishaan Verma      | - 23BCE11052 |
| Harshit Arora     | - 23BCE10911 |
| Aashish Pandey    | - 23BCE11066 |
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# OBJECTIVE

**Better Applications  
Brighter Opportunities**

01

## MOTIVATION

Job seekers manage applications, resumes, interview preparation, follow-ups and skill requirements across multiple disconnected tools.

02

## AIM

To build a unified AI-powered platform that combines application tracking with candidate profiling, job matching and personalized career intelligence.

03

## OBJECTIVES

- Centralize job applications, documents and reminders.
- Parse resumes and build a structured candidate profile.
- Match candidate skills and experience with job requirements.
- Provide an explainable six-factor application-readiness score.
- Track evidence behind candidate skills through Skill Proof.
- Maintain a Career Twin for personalized career guidance.
- Recommend the Next Best Action for each application.
- Support cover-letter generation and interview preparation.

# PHASE-1 REVIEW → PHASE-2 REMEDIAL MEASURES

| Review-1 Focus / Suggestion            | Phase-2 Remedial Measure                                           | Current Status |
|----------------------------------------|--------------------------------------------------------------------|----------------|
| Complete AI integration                | Integrated Gemini-powered career intelligence workflows            | ✔ Implemented  |
| Expand beyond basic tracker            | Added Match, Skill Proof, Career Twin and Next-Best-Action         | ✔ Implemented  |
| Improve calendar/tracker functionality | Interview & follow-up tracking integrated                          | ✔ Implemented  |
| Strengthen authentication              | Firebase ID-token verification with user-scoped backend routes     | ✔ Implemented  |
| Improve document handling              | Backend document APIs + 5 MB + MIME + content-signature validation | ✔ Implemented  |
| Improve application workflow           | Duplicate protection and Saved → Applied transition                | ✔ Implemented  |
| Improve deployment architecture        | Vercel frontend + Render backend configuration                     | ✔ Configured   |
| Improve security                       | Backend-only Gemini credentials and restricted production API flow | ✔ Implemented  |
| Improve explainability                 | Six-factor readiness scoring + Skill Proof + Career Twin           | ✔ Implemented  |

Phase 1 built the tracking foundation. Phase 2 turned it into an integrated career-intelligence platform.

# EXISTING PROJECTS

| Capability            | Huntr                 | Teal                 | Simplify      | JobPilot                                   |
|-----------------------|-----------------------|----------------------|---------------|--------------------------------------------|
| Core focus            | Visual pipeline / CRM | Resume & ATS scoring | Autofill      | Application tracking + Career Intelligence |
| Resume AI             | Basic tailoring       | ATS matching         | Limited       | Gemini-assisted parsing                    |
| Application tracking  | ✓                     | ✓                    | Limited       | ✓                                          |
| Document management   | Limited               | Limited              | No            | Per-application documents                  |
| Skill evidence        | Limited               | Limited              | No            | Skill Proof                                |
| Career intelligence   | Limited               | Limited              | No            | Career Twin + NBA                          |
| Readiness assessment  | Limited               | Match/ATS            | No            | Six-factor explainable score               |
| Interview preparation | Limited               | Limited              | Limited       | AI-assisted preparation                    |
| Data ownership        | Vendor-hosted         | Vendor-hosted        | Vendor-hosted | User-scoped backend data                   |

**Key Differentiator: JobPilot connects application tracking with evidence-based career intelligence instead of separate workflows.**

# PROJECT MODULES

01

## Application Management

Track Saved → Applied → Under Review → Interview → Offer / Rejected.

03

## Job Matching

Compare candidate skills, projects and experience against job requirements.

05

## Skill Proof

Measure the evidence strength behind individual skills.

07

## Next-Best-Action Coach

Recommend the highest-priority action for improving readiness.

09

## Documents & Reminders

Manage application documents and follow-up/interview reminders.

02

## Resume & Candidate Profile

Extract resume information and maintain a structured candidate profile.

04

## Application Success Predictor

Explain application readiness using six weighted factors.

06

## Career Twin

Structured representation of strengths, weaknesses and interview areas.

08

## Interview & Career Preparation

Generate role-specific preparation and interview assistance.

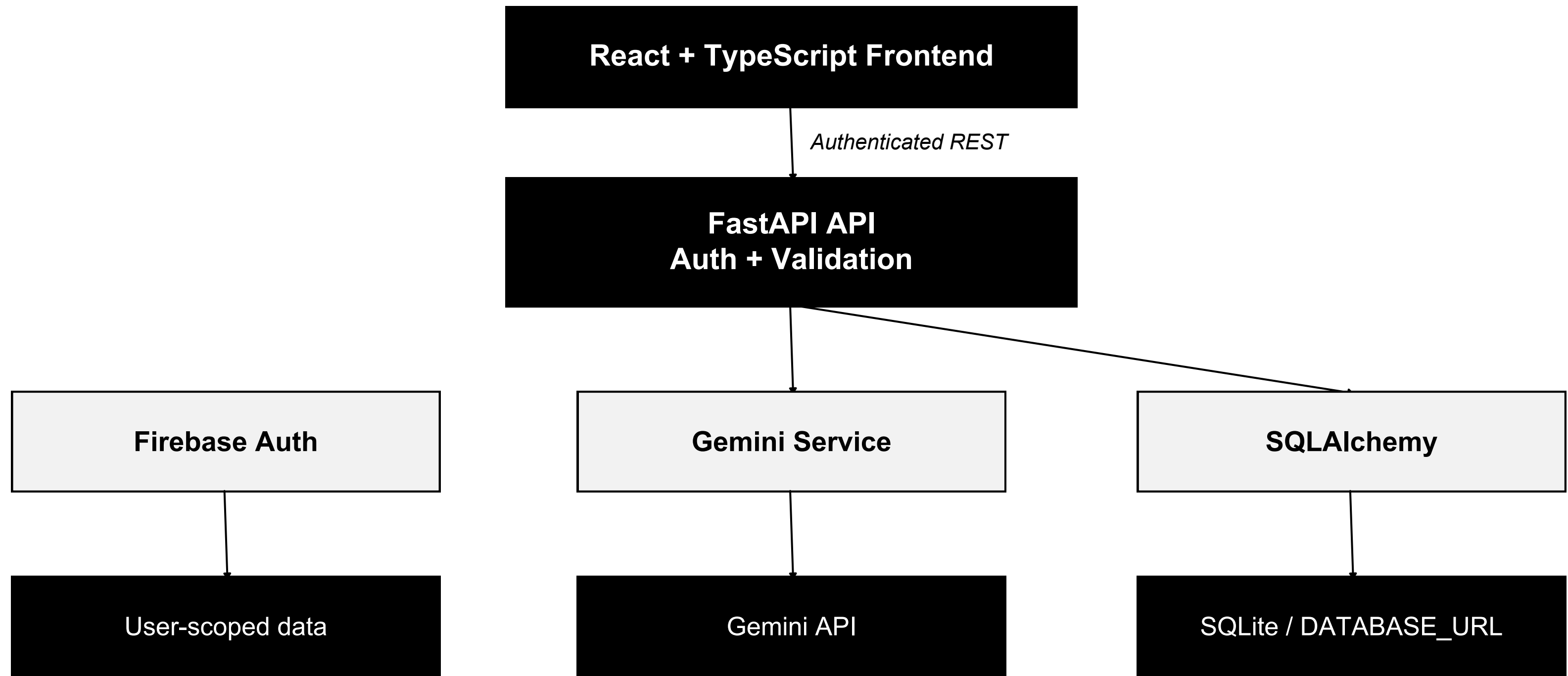
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## Authentication & Secure API

Firebase authentication + user-scoped FastAPI operations.



# SYSTEM ARCHITECTURE



AI credentials remain backend-side; the browser never directly accesses the Gemini provider.

# CAREER INTELLIGENCE WORKFLOW



**Match**

How relevant is the candidate to the role?

**Prove**

How strong is the evidence behind each skill?

**Predict**

What is the candidate's application readiness?

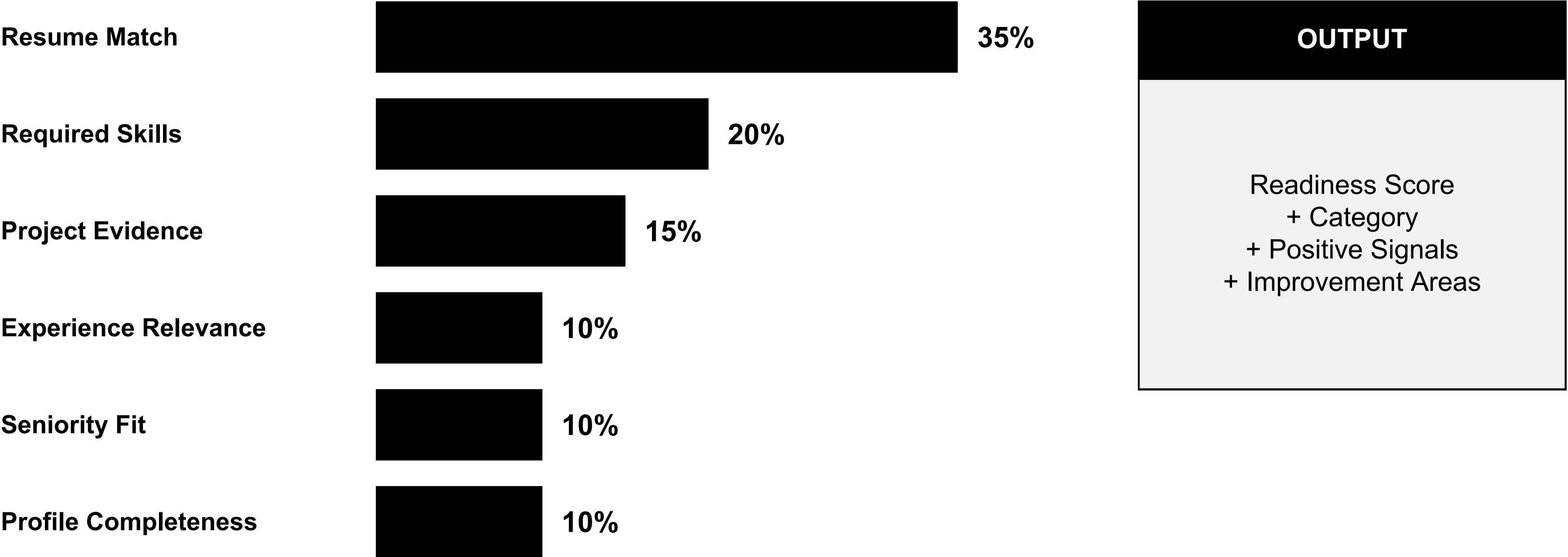
**Act**

What should the candidate do next?

**Improve**

How does the candidate become stronger for future opportunities?

# EXPLAINABLE APPLICATION READINESS



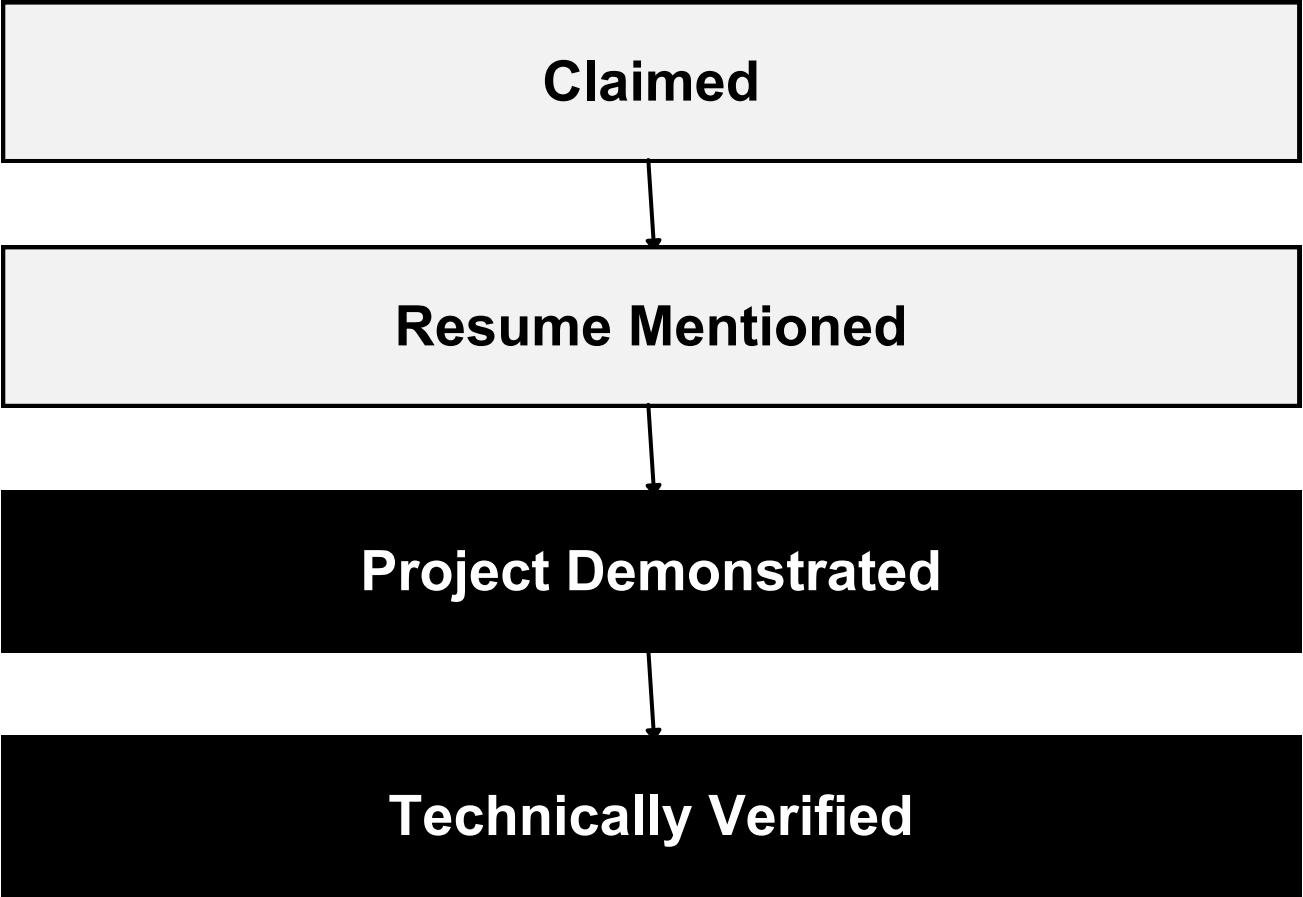
*Total = 100%*

Project-defined readiness heuristic, not a statistically calibrated probability of interview or offer.



# EVIDENCE-BASED CAREER INTELLIGENCE

## Skill Proof



Measures the strength of evidence behind a candidate's skills.

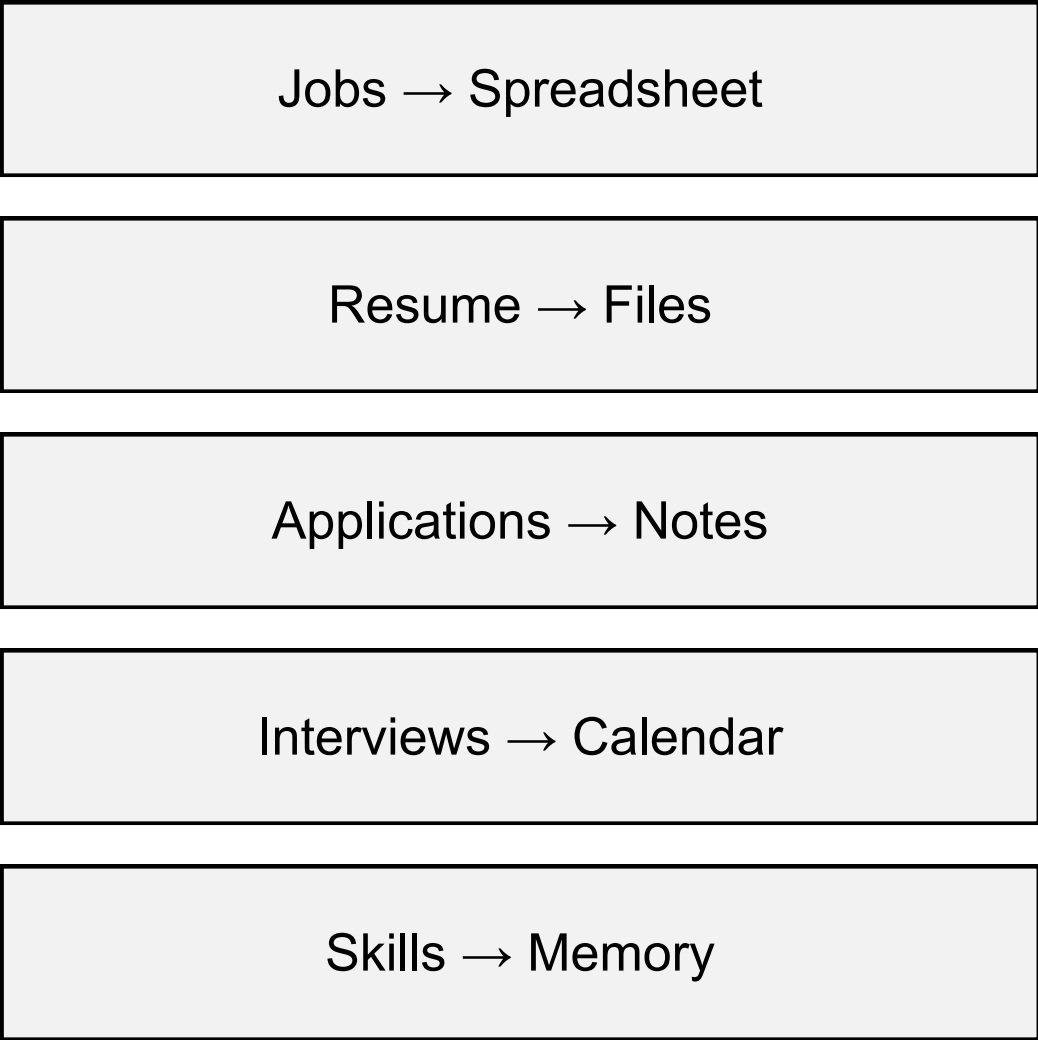
## Career Twin

- Maintains:
- Skills
  - Projects
  - Experience
  - Strengths
  - Weaknesses
  - Interview areas
  - Performance notes

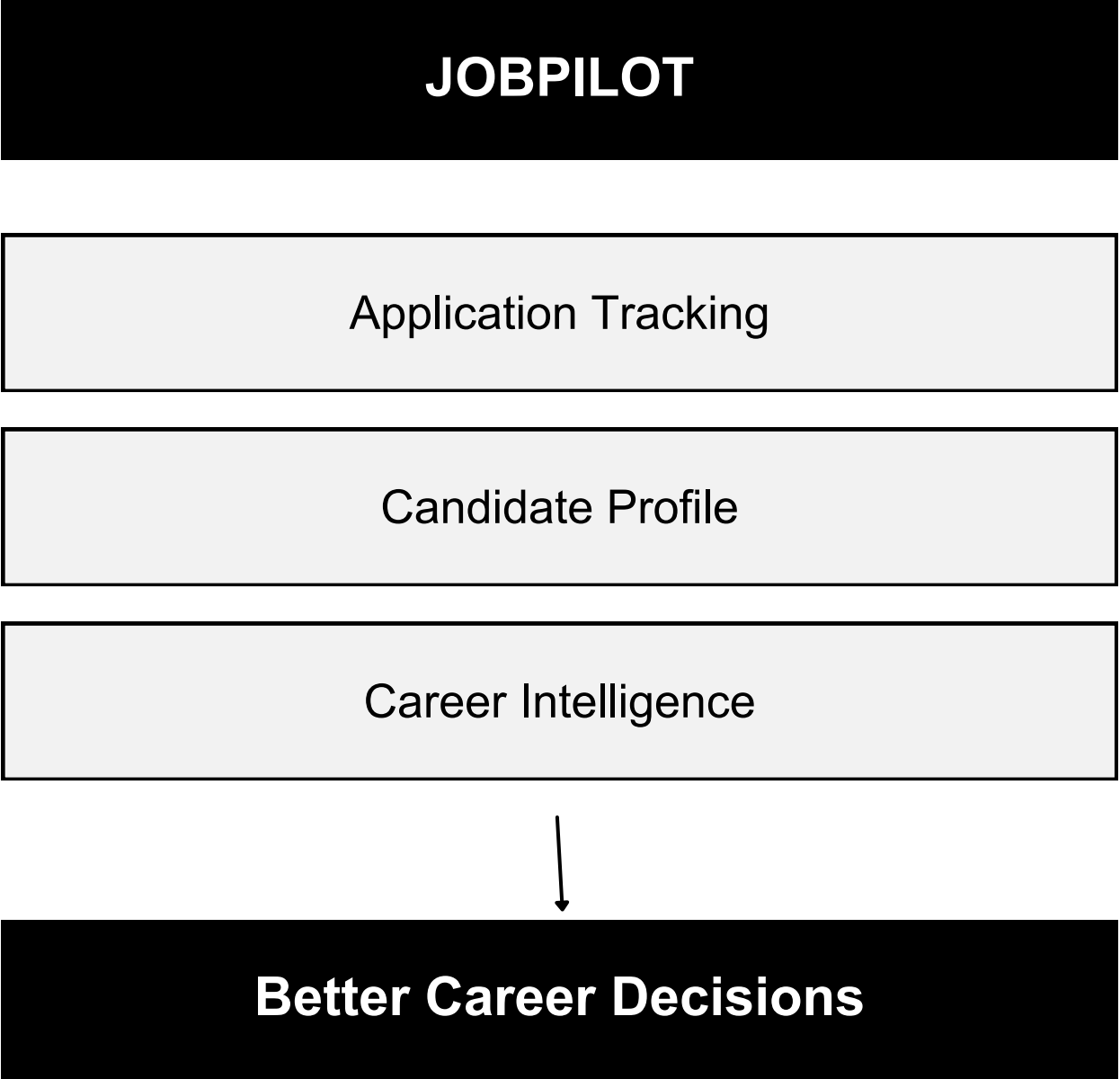
Move from “I have this skill” to “I can prove this skill.”

# PROJECT OUTCOME

## Before JobPilot



## With JobPilot



## Community Impact

- Reduces fragmented job-search management
- Helps candidates understand job-fit factors
- Makes skill gaps visible
- Provides actionable career guidance
- Centralizes applications, documents and follow-ups
- Supports interview preparation

# INDIVIDUAL CONTRIBUTIONS

## Harshit Arora

- React/Vite frontend
- Dashboard & job browsing
- Application tracker
- Profile/resume interfaces
- Career Intelligence UI
- Responsive UI
- Frontend-backend integration

## Ishaan Verma

- FastAPI backend
- REST APIs
- Authentication integration
- Application/job/profile APIs
- Database communication
- Backend error handling

## Aashish Pandey

- AI & Career Intelligence
- Resume/job matching
- Application Success Predictor
- Skill Proof
- Career Twin
- Interview preparation
- AI recommendations

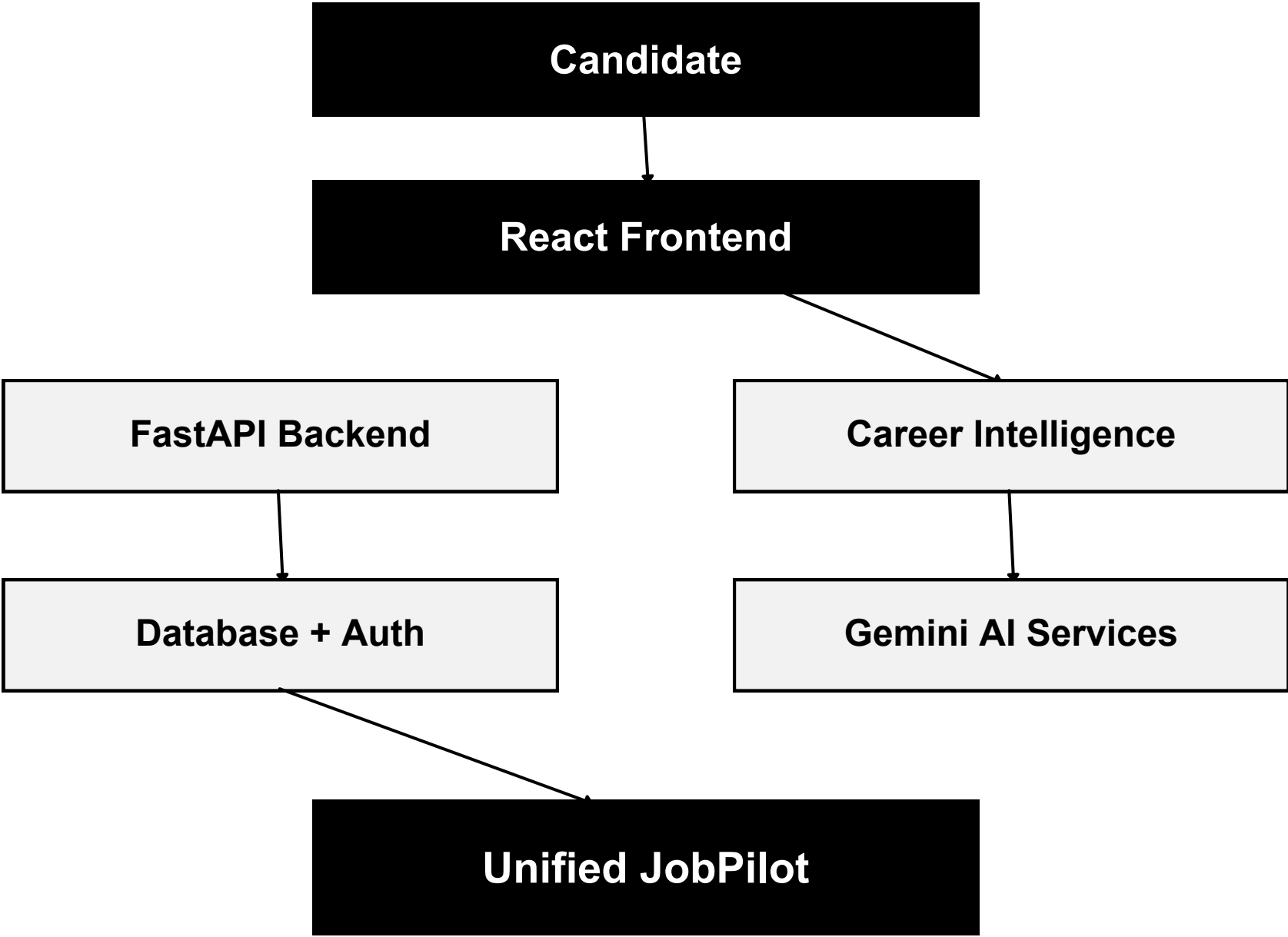
## Udit Narayan Rout

- Database architecture
- Data persistence
- Authentication/data management
- CRUD operations
- User-scoped storage/retrieval

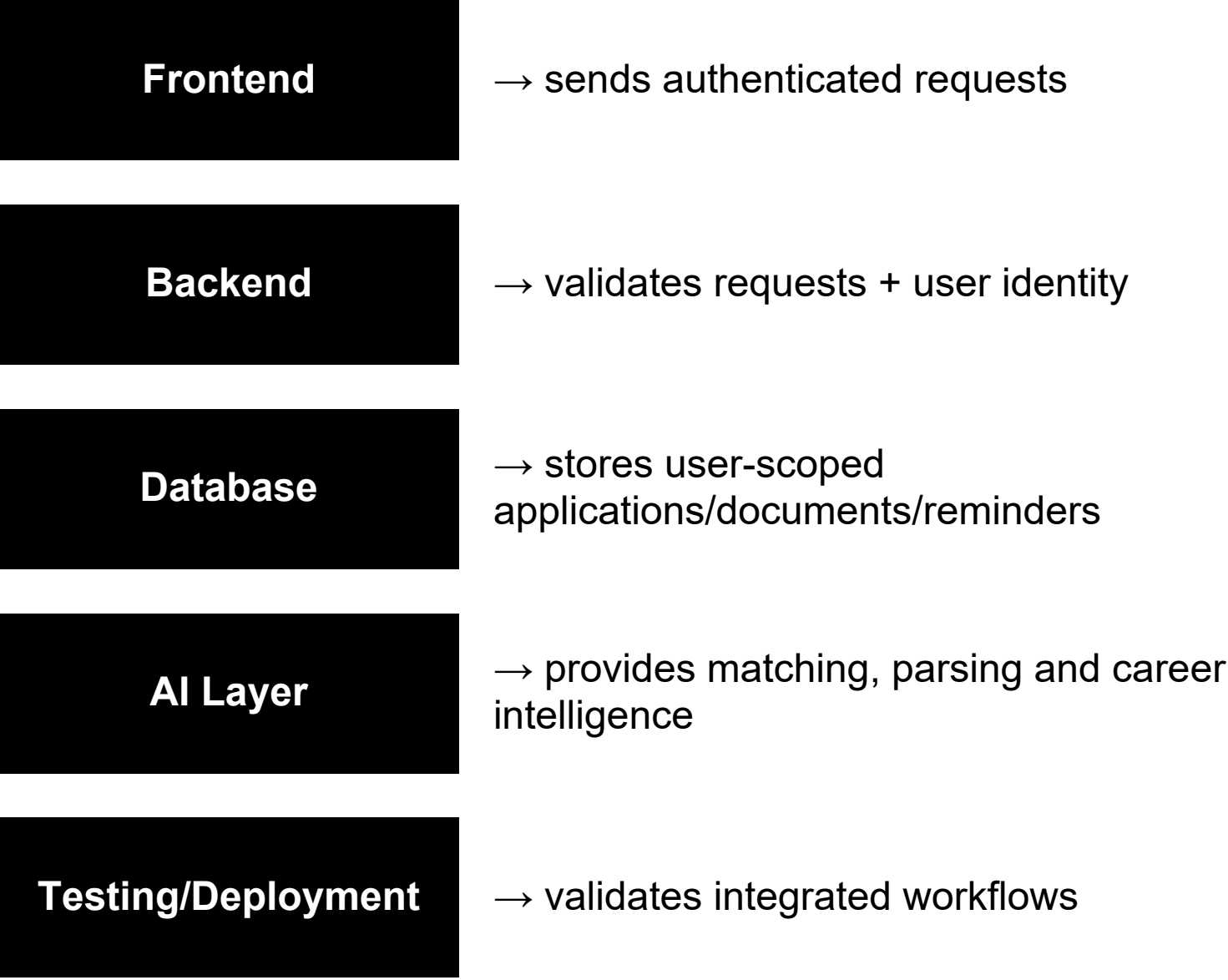
## Kartikey Tiwari

- Testing
- API/integration testing
- Deployment
- Vercel/Render configuration
- Health checks
- End-to-end validation

# WORK INTEGRATION



## Integration flow



# PHASE-2 IMPLEMENTATION STATUS

| Module                        | Status        |
|-------------------------------|---------------|
| Application Management        | ✔ Implemented |
| Resume/Profile Processing     | ✔ Implemented |
| AI Job Matching               | ✔ Implemented |
| Application Success Predictor | ✔ Implemented |
| Skill Proof                   | ✔ Implemented |
| Career Twin                   | ✔ Implemented |
| Next-Best-Action Coach        | ✔ Implemented |
| Interview Preparation         | ✔ Implemented |
| Documents & Reminders         | ✔ Implemented |
| Firebase Authentication       | ✔ Implemented |
| Backend Gemini Gateway        | ✔ Implemented |
| Deployment Configuration      | ✔ Implemented |

### LIMITATIONS

- Job catalogue remains curated/static
- Recruiter inbox uses sample data
- External applications are not submitted automatically
- Live deployment/test results are claimed only if actually verified



# FUTURE SCOPE

## Phase 3 — Real-Time Career Intelligence

- Live authorized job-provider integrations
- Job verification & expiry detection
- Job deduplication
- Job-specific resume optimization
- Evidence-based skill graph
- Advanced AI interview simulator
- Enhanced Career Twin

## Phase 4 — AI Career Agent

- Personalized career goals
- Continuous Next-Best-Action recommendations
- Application outcome learning
- Candidate-specific opportunity ranking
- Career market intelligence
- Continuous candidate improvement loop

**Long-term Vision: evolve JobPilot from an application tracker into an AI-powered career operating system.**

# REFERENCES

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- [Zod Documentation](#)
- [Gemini API Documentation](#)
- [FastAPI Documentation](#)
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- [Firebase Admin SDK — ID Token Verification](#)
- [TanStack Router Documentation](#)
- [Vite Documentation](#)
- [React Documentation](#)

**THANK YOU**